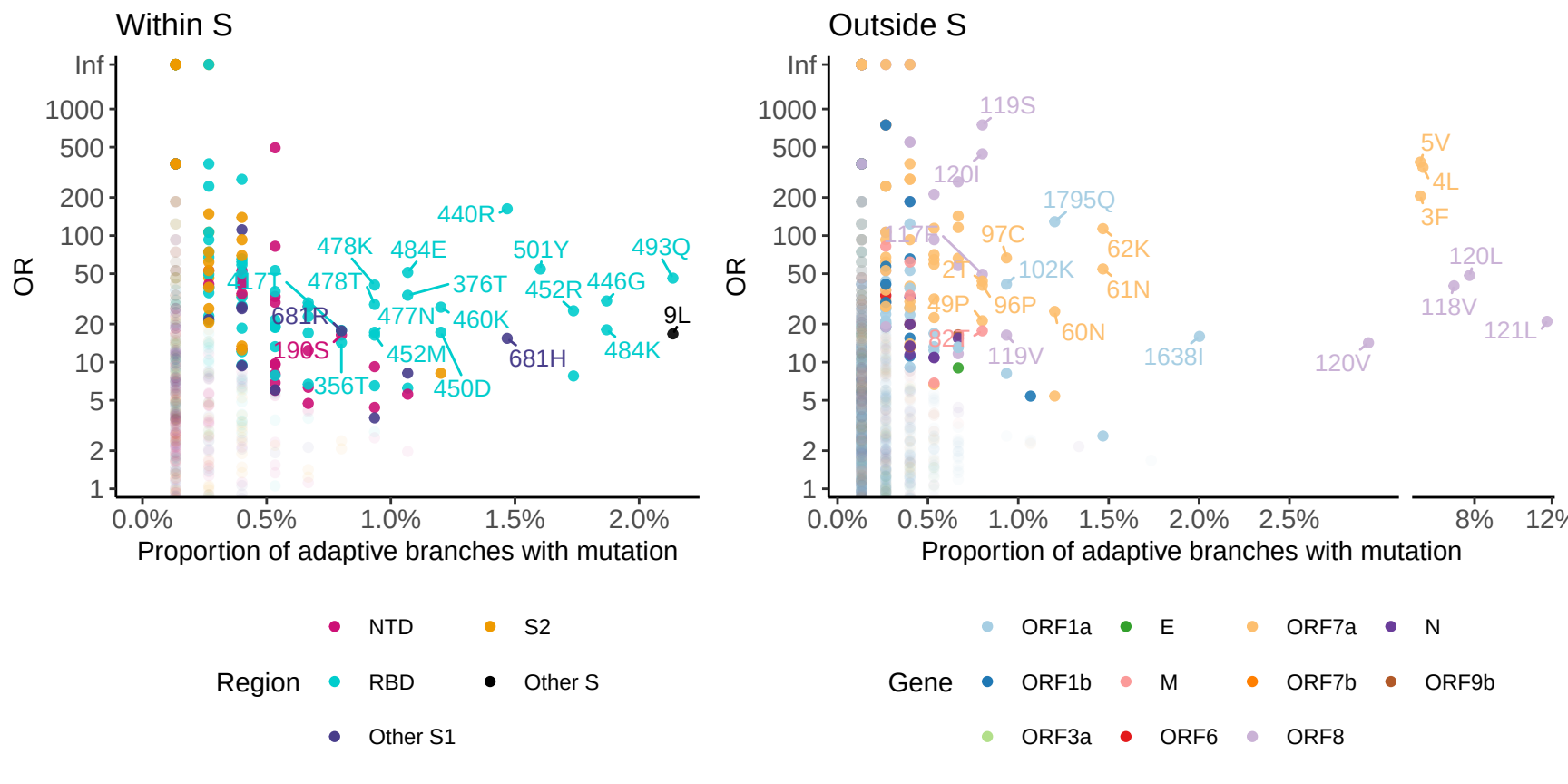

Within S
Outside S
OR
Inf
1000
500
200
100
50
20
10
5
2
1
0.0%
0.5%
1.0%
1.5%
2.0%
2.5%
8%
12%
Proportion of adaptive branches with mutation
440R
478K
484E
501Y
493Q
417T
478T
376T
452R
446G
681R
477N
460K
9L
190S
452M
681H
484K
356T
450D
119S
120I
1795Q
5V
4L
3F
120L
118V
121L
97C
62K
102K
61N
49P
96P
60N
119V
1638I
120V
Region
NTD
S2
RBD
Other S
Other S1
Gene
ORF1a
E
ORF7a
N
ORF1b
M
ORF7b
ORF9b
ORF3a
ORF6
ORF8